AIME 2025 Presentation Talking Points

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Slide 1 — Reflection

We can think of, say, Alexander the Great being tutored by Aristotle. Part of what makes tutoring so powerful is personalized reflection. The learner is being encouraged to critically work with their thoughts.

LLMs are capable of dialogue, given this we should see LLM-guided tutoring should enhance learning by enabling reflection.

Slide 4 — Technical Details

Talk through the diagram, explain how the system works with LiveKit agents, real-time audio processing, and how the reflection prompts are integrated into the podcast flow.

Slide 6 — Details (Study Design Diagram)

Attractiveness in the UEQ specifically measures the overall impression of the product - whether users like or dislike it. It's a pure valence dimension that captures the general appeal of the system to users.

Slide 8 — Procedure

The reflection prompts specifically asked: **"What was the most important thing you learned?"** This was designed to encourage metacognitive reflection, but in practice it may have focused too much on factual recall rather than deeper synthesis or goal-oriented thinking.

User Experience Questionnaire (UEQ) - This is a standardized measure for assessing user experience across multiple dimensions including attractiveness, perspicuity, efficiency, dependability, stimulation, and novelty. Similar to how other VR studies measure user experience (see Khan et al. (Omar's paper!), 2024 on avatar-locomotion congruence in VR), we used the UEQ to capture subjective user perceptions.

Slide 10 — Some Numbers (Attractiveness Plot)

- Standard condition had higher attractiveness ratings: M=29.56 (SD=4.00)
- Reflection condition had lower attractiveness: M=26.22 (SD=4.58)
- This was statistically significant (t=2.26, p=0.03, D=0.75) with a medium-to-large effect size.

Users in the reflection condition complained that the reflection prompts disrupted their flow and felt intrusive, which explains the lower attractiveness scores.

Slide 11 — Possible Explanations and Lessons

Why did reflection fail?

- We did factual recall ("what did you learn?") instead of something natural (like the goal-oriented synthesis in the Cloude paper)

- We had a binary evaluation (understood/didn't understand) which was too coarse, and didn't capture depth

- **Unfortunate Timing!** We prompted at end of sections rather than when learner actually needed it, because we can't detect that

- **Agency!** We also forced engagement at AI's discretion, which wasn't learner driven

- We weren't building a profile of the learner or keeping in mind their goals, or examining their metacognitive state

This was more interrupt-y than support-y. The standard condition let learners interrupt freely when they needed help. The reflection condition forced engagement at arbitrary times. The data suggests learners may know when they need help... **maybe the best AI tutor knows when to stay silent.!!!** (key point)

Lessons learned

- Try to understand the learner and have good timing

- Agency > forcing the user to do something

- Maybe adapt to engagement signals using multimodal data
- Align prompts with learner goals (tricky!!!)

Slide 12 — Future Work

Why more data helps with reflection timing: Currently, we prompted reflection at fixed intervals (end of sections). But learners have different needs - some might benefit from reflection when confused, others when making connections, still others when losing focus.

More data (eye tracking, physiological signals like heart rate variability, facial expressions, interaction patterns) could help us detect the learner's cognitive and emotional state in real-time. This would allow us to prompt reflection at moments when learners actually need it - when they're confused, overwhelmed, or disengaged - rather than at arbitrary points.

This could transform reflection from an interruption into adaptive support that respects learner agency while still encouraging metacognition at the right moments.

"Maybe learners know when they need help" - but how to judge when they need prompting? Maybe not that simple - might need to detect micro facial expressions or other subtle signals. This takes dedicated effort. Not that AI can't ever be a tutor... but we need tutors that are **intentional!!!!!!** about how they teach. Further research is necessary.

Specific future directions

- Track prior reflection quality over time to build learner models

- Test different prompt types: planning prompts, strategy prompts, comparison prompts - not just recall
- Move beyond binary assessment (understood/didn't understand) to reflection depth rubrics that capture nuance
- Investigate when learners actually want to reflect vs when they feel interrupted